

CoreMetrics

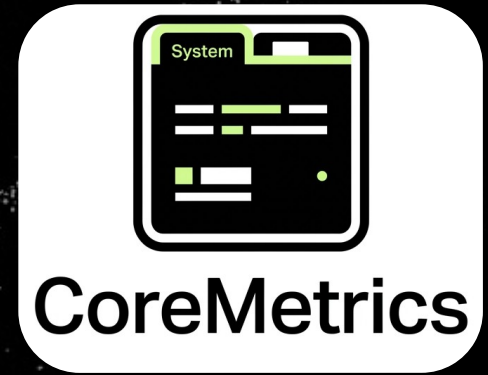


Real-time system monitor
C++23 · SDL3 · cross-platform

Alicia Melotik

Stefan
Oleksiienko

Martin Castellanos





What we will cover

- 1 Product walkthrough
- 2 Architecture
- 3 Cross-platform proof
- 4 Engineering features
- 5 Live demo + Q&A

What we built

- GUI library + demo app
- Two tabs, live metrics
- 80+ unit tests, 11 suites
- Real /proc, IOKit, PDH
- Click, sort, exit, mute



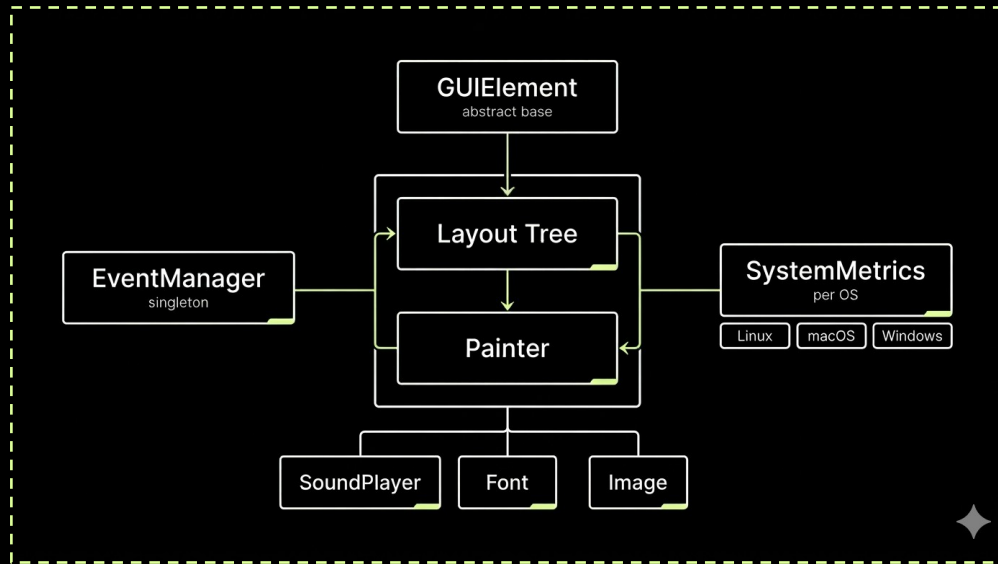
User story

- Dev notices slowdown
- Opens CoreMetrics
- Spots CPU red bar
- Switches to Processes tab
- Identifies offending PID



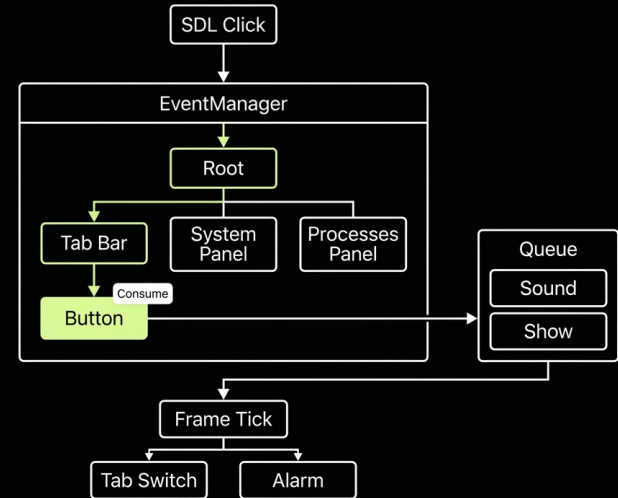
Architecture overview

- GUIElement abstract base
- Layout tree, painter render
- EventManager singleton
- SystemMetrics per OS
- SoundPlayer, Font, Image



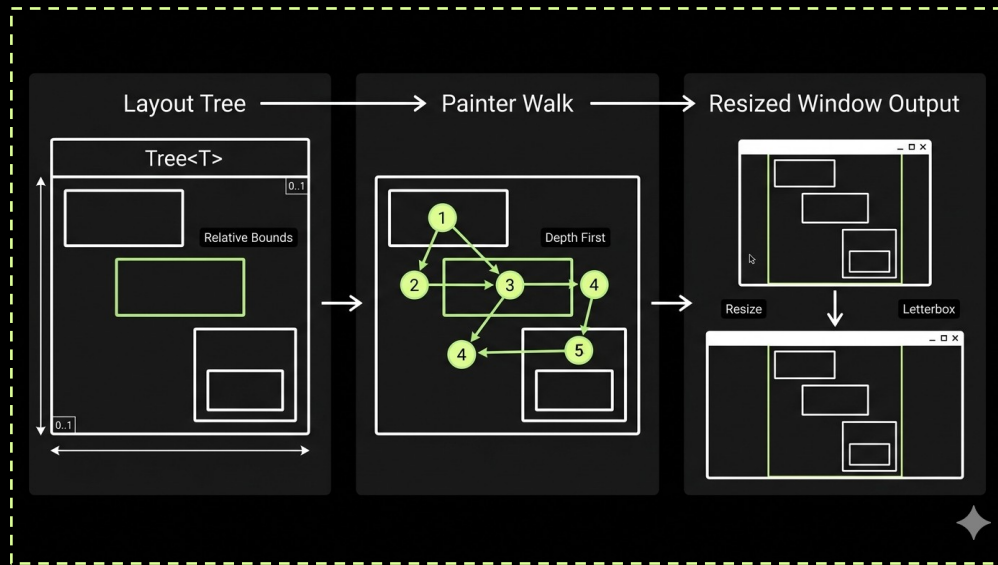
Event system

- Trickle propagation
- Functor on every element
- Click, Show, Sound events
- Queue drained per frame
- Tab switch, alarm fire



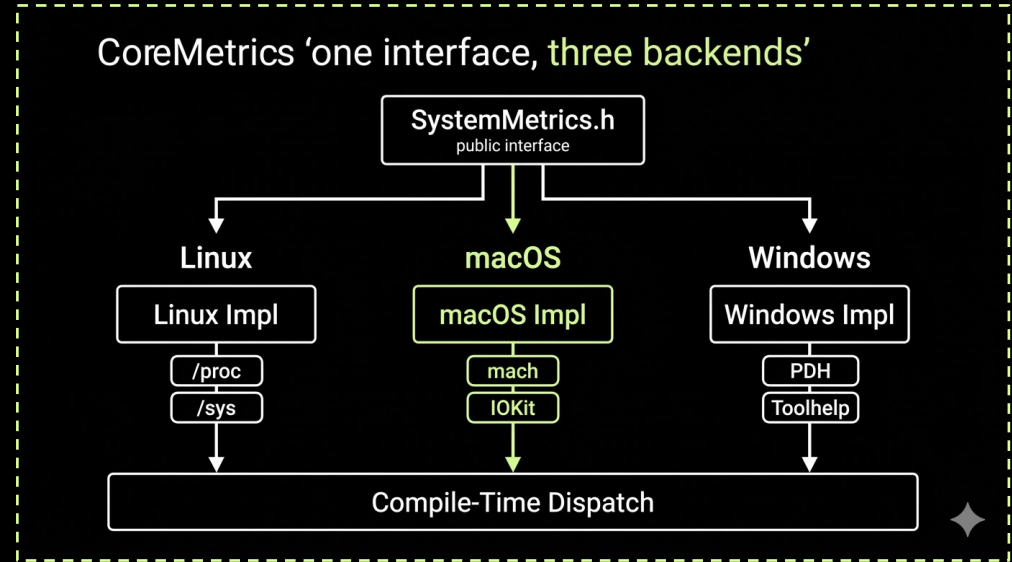
Layouts and rendering

- `Tree<T>` generic container
- Layout = relative bounds
- Painter algorithm walk
- Aspect-preserving blit
- Window resize handled



Cross-platform metrics

- Linux: /proc + /sys
- macOS: mach + IOKit
- Windows: PDH + Toolhelp
- One header, three impls
- Compile-time dispatch



Rubric: C++ features

- CRTP via Cloneable<Derived>
- Covariant return clone
- Move semantics throughout
- Templates: Tree, Cloneable
- Singletons (4) + factory



Rubric: STL and patterns

- Containers: vector, queue, map
- Iterators in GUIFile parser
- Functors: operator() on element
- Algorithms: std::sort
- File I/O: XML, /proc, WAV

✓
Containers
vector
queue
map

✓
Iterators
GUIFile
parser
XML walk

✓
Functors
operator()
events

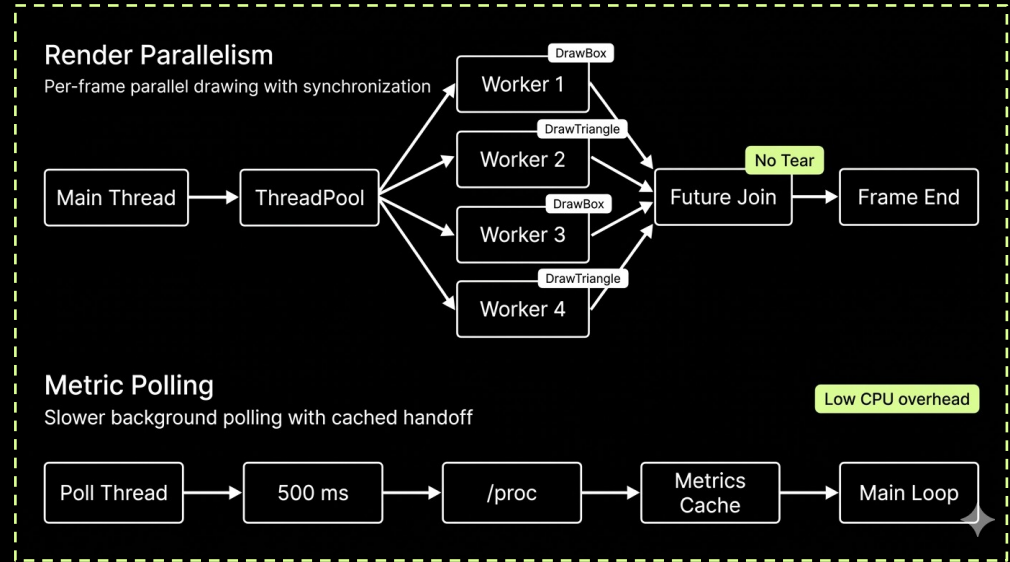
✓
Algorithms
std::sort
process list

✓
File I/O
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/proc
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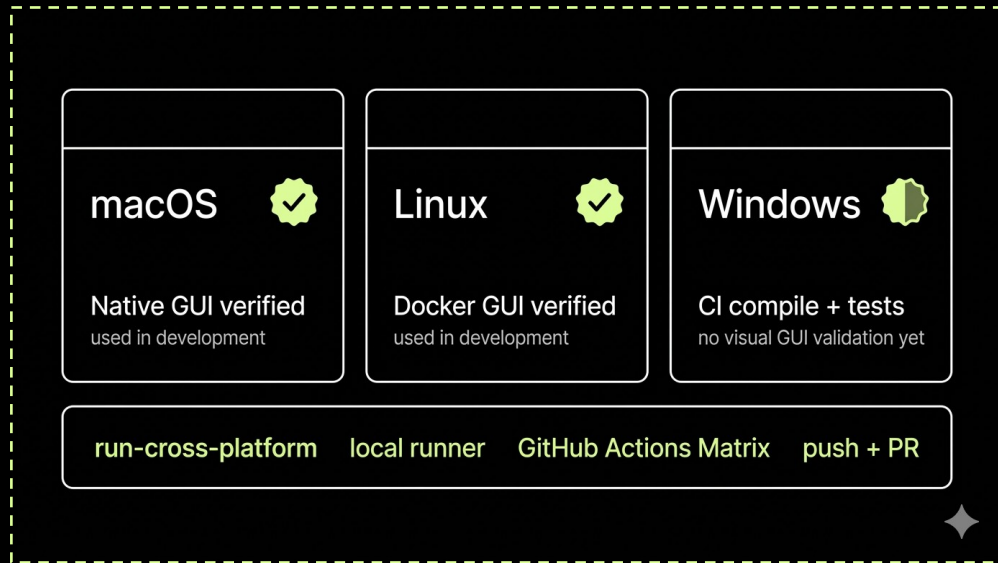
Concurrency

- ThreadPool singleton
- DrawBox parallelized rows
- DrawTriangle parallelized
- Future-based join per frame
- 500ms metric poll thread



Cross-platform proof

- macOS native, verified
- Linux Docker, verified
- Windows CI, compile + tests
- One script: run-cross-platform
- GitHub Actions matrix



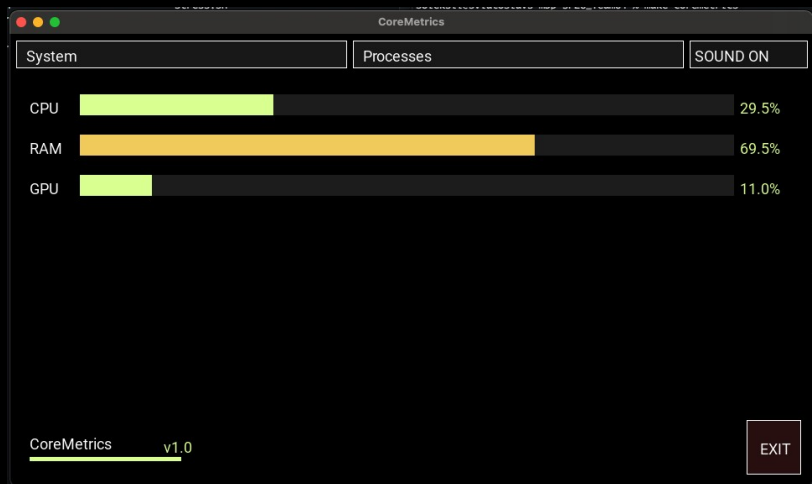
Stress test

- stress.sh spikes load
- CPU workers + RAM alloc
- GPU via glmark2 / WebGL
- Bars cross thresholds
- Alarm fires, mute toggle



CoreMetrics in action

Live demo

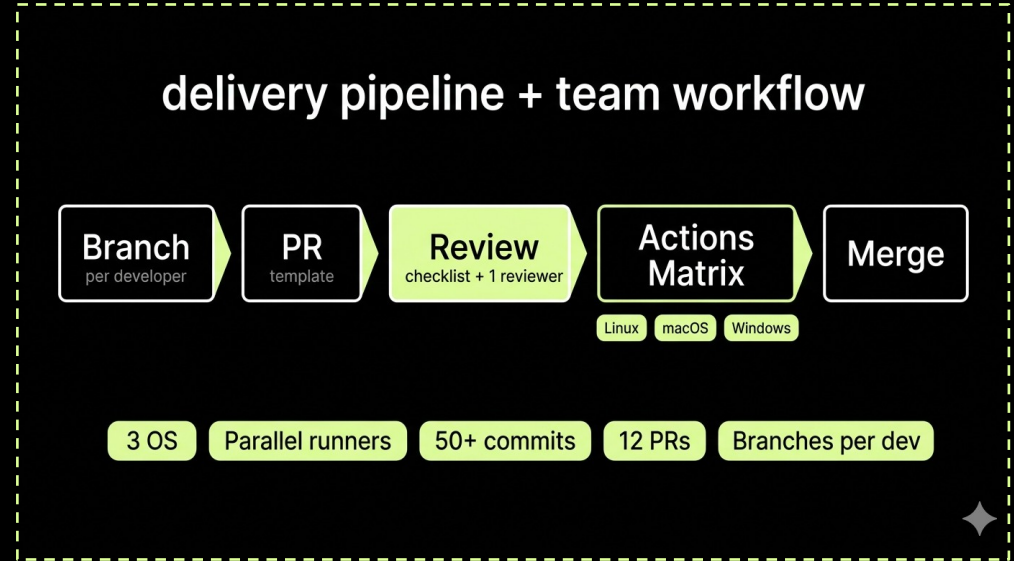


The CoreMetrics v1.0 application window is shown with the 'Processes' tab selected. It displays a table of running processes, including their PID, Name, CPU%, and MEM%. The interface includes a 'SOUND ON' button and an 'EXIT' button at the bottom right.

PID	NAME	CPU%	MEM%
24617	plugin-container	0.5	5.3
4645	plugin-container	0.2	3.5
24562	plugin-container	0.3	3.1
62732	plugin-container	0.0	2.6
62733	plugin-container	0.0	2.2
25510	plugin-container	0.1	2.1
4607	firefox	0.1	2.0
4629	plugin-container	0.0	1.7
90802	com.apple.Virtualization.Virtua	0.0	1.0
2177	Slack Helper (Renderer)	0.0	1.0
34479	Microsoft PowerPoint	0.0	1.0
32867	plugin-container	0.0	0.8
10502	plugin-container	0.0	0.8
29685	plugin-container	0.0	0.8
35250	plugin-container	0.0	0.8

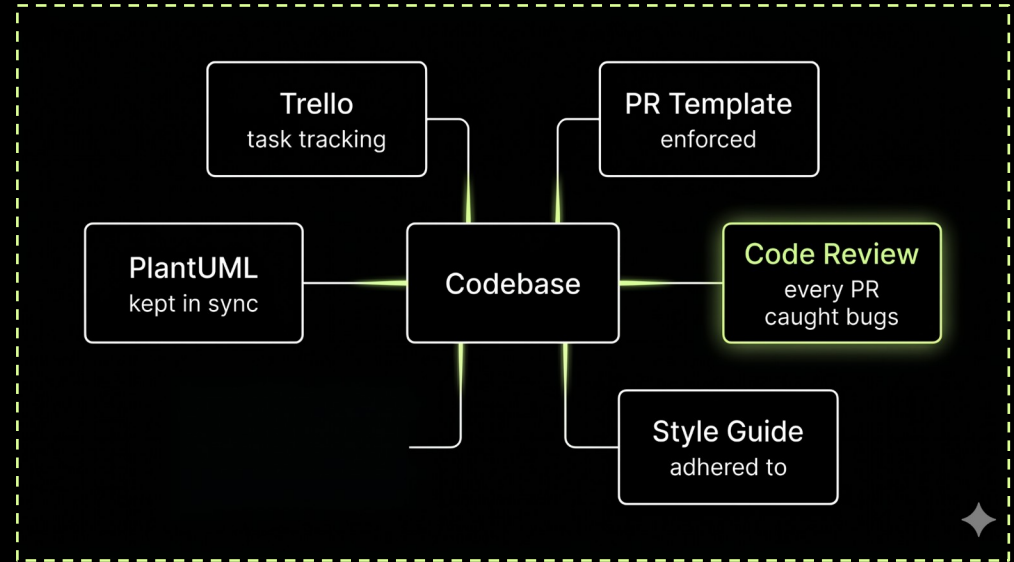
CI / DevOps

- GitHub Actions matrix
- 3 OS, parallel runners
- PR template + reviews
- 50+ commits, 12 PRs
- Branches per developer



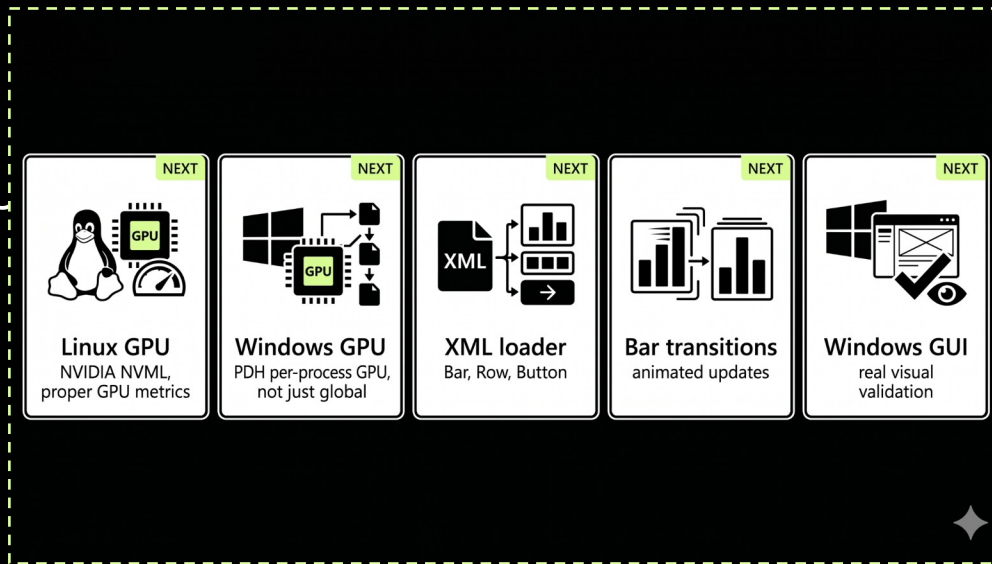
Engineering process

- Trello task tracking
- PR template enforcement
- Code review per PR
- Style guide adhered
- PlantUML diagrams maintained

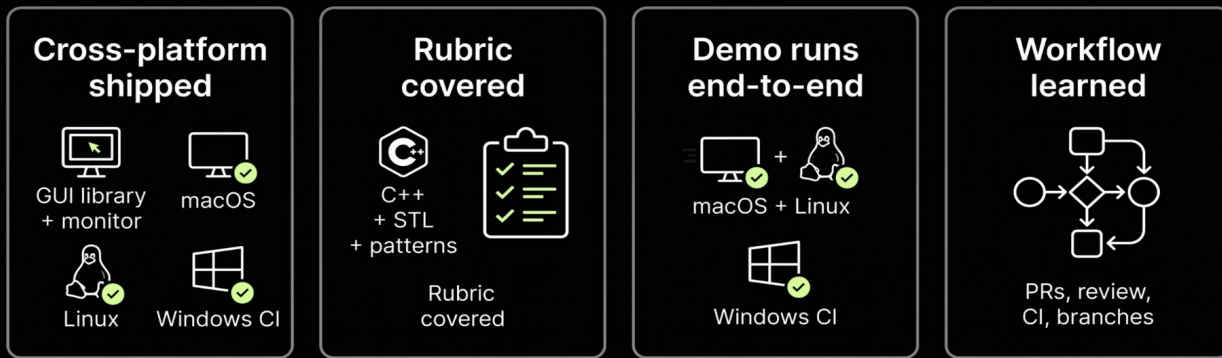


What we would do next

- NVIDIA NVML for Linux GPU
- Windows PDH per-process GPU
- XML parses bar/row/button
- Animated bar transitions
- Real Windows visual test



Closing



Questions?

